

Why IPV6?

- IPV4 are Limited
- Total IP= 4,294,967,296
- NAT and VLSM are Short Term Solution.
- Assigned by IANA
- On Sept 2015 ARIN declared that they have exhausted ARIN IPV4 Pool.
- On Aug 2020 LACNIC Announced that they allotted Last IP
- IPV4 is in Decimal Format but IPV6 is in Hexadecimal Format.

Hexadecimal Number System

Binary: Base 2

Numbers: 0 or 1

Decimal Number: Base 10

Numbers: 0,1,2,3,4,5,6,7,8,9

Hexadecimal Number: Base 10

Numbers: 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F

Each digit has 4 Bits

Hexadecimal Numbers:

0000 – 0

0001 – 1

0010 – 2

0011 – 3

0100 – 4

0101 – 5

0110 – 6

0111 – 7

1000 – 8

1001 – 9

1010 – A

1011 – B

1100 – C

1101 – D

1110 – E

1111 – F

IPV6:

- IPV6 is 128 Bits of addresses.
- IPV6 is having 8 blocks of 16 Bits.
- total 340 Trillion Trillion Trillion Addresses.
- Written in Hexadecimal

Example:

2001:0DB8:5917:EABD:6562:17EA:C92D:09BD

As we can see that the IPV6 is longer address so we have methods to shorten the IPV6 Addresses.

Method to Shorten IPV6 Address:

1. Remove Preceding Zeroes

Eg: 2001:0DB8:000A:001B:20A1:0020:0080:34BD

Remove Marked Zeros

2001:**0**DB8:**000**A:**00**1B:20A1:**00**20:**00**80:34BD

2001:DB8:A:1B:20A1:20:80:34BD

2. Remove Consecutive Zeroes:

Eg: 2001:**0**DB8:**0000:0000:0000:0000:00**80:34DB

Ans: 2001:DB8::**80:34DB**

IPV6 Prefix:

1. If an enterprise request the IPV6 from ISP, it will get a /48 Block.
2. Typically IPV6 uses a /64 Block.
3. That means an enterprise has 16 Bits to make subnets.
4. The remaining bits can be used for host.

Eg: 2001:0DB8:8B00:0001:0000:0000:0000:0001/64

48 Bits of Global
routing Prefix
assigned by ISP

16 Bits
of
Subnets

64 Bit of interface Identifier, host
portion of the address.

Eg: 300D:00F2:0B34:2100:0000:0000:1200:0001/56

52 Bits

56 Bits

Find Network

Eg: 2001:0DB8:8B00:0001:FB89:017B:0020:0001/93

16

32

48

64

80

84

88

92

93+3

2001:0DB8:8B00:0001:FB89:017**B**:0020:0001/93

1011

1000

Ans: -

2001:0DB8:8B00:0001:FB89:0178:0000:0000/93

2001:0DB8:8B00:0001:FB89:0178::/93

EUI Identifier:

- Stands for Extended Unique Identifier
- Modified EUI 64 method of Converting y the MAC 48 Bits into the IP Address of 64 Bit Interface Identifier.
- This interface identifier can become the host portion of a /64 IPV6 Address

Eg: Convert 1234 5678 9ABC to EUI-64

Insert FF:FE in Between the MAC Address

1234 56 **FF FE** 78 9ABC Invert 7th Bit


4 Bit 8 Bit

0010 → Invert the Bit → 0000

1034 56 **FF FE** 78 9ABC - Host Part

Global Unicast Address: -

- IPV6 Address that can be used over the internet.
- Must register, because they are public address, it is expected that they are globally unique.
- Originally Defined as the 2000:: /3

- Range:

2000:0000:0000:0000:0000:0000:0000:0000/3

To

3FFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/3

- Defined as all the address not reused for other purpose

2000 – 0010 0000 0000 0000

/3 means **First 3 fix** and then we can increase to one

Then Convert to Decimal

0011 0000 0000 0000 → 3000::/3

0011 0000 0000 0000 → 3FFF::/3

Unicast Local Address:

- They Cannot be used over the internet.
- IPv6 Private IP Address.
- Don't need to register them.
- Can be used in internal networks.
- Cannot be routed over internet.

Range:

FC00:0000:0000:0000:0000:0000:0000:0000/7

To

FDFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/7

FC00 – 1111 1100 0000 0000

/7 means **First 7 Bits fix** and then we can increase to one

Then Convert to Decimal

1111 1101 0000 0000 → FD00::/7

1111 1101 1111 1111 → FDFF::/7

Link Local Address:

- Automatically Generated on IPV6 Enabled Interfaces.
- Used command R1(Config)#ipv6 enable on an interface.
- Used the address block FE80::/10.
- Can be seen as FE9, FEA, FEB.
- It is generated as the EUI - 64 Rules.
- Will not Route Packets, not used for Communication.

Range:

FE80::/10 or

FE80:0000:0000:0000:0000:0000:0000:0000/10

To

FEBF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/10

FE80 → 1111 1110 1000 0000

/10 means **First 10 Bits fix** and then can be increased

Then Convert to Decimal

1111 1110 1000 0000 → **FE80::/10**

1111 1110 1001 0000 → **FE90::/10**

1111 1110 1010 0000 → **FEA0::/10**

1111 1110 1011 0000 → **FEB0::/10**

1111 1110 1011 0000 → **FEBF::/10**

Multicast Address:

- Unicast address used for one to one communication are source to one destination.
- Broadcast address are one to all.
→ One Source to all destination.
- Multicast address are one to many.
→ One Source to multiple destination or specific group.
- IPv6 Uses the FF00::/8 for the multicast.

Range:

FF00::/8

To

FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/8

Multicast Alloted Address:

Purpose:

**All Nodes/Host
Like Broadcast**

IPV6 Address

FF02::1

IPV4

224.0.0.1

All Routers

FF02::2

224.0.0.2

All OSPF Routes

FF02::5

224.0.0.5

**All OSPF
DRs/BDR**

FF02::6

224.0.0.6

ALL RIP

FF02::9

224.0.0.9

ALL EIGRP

FF02::10

224.0.0.10

Anycast Address:

- Anycast address is a set of address that is assigned to a set of interfaces typically belong to different nodes. A Packet sent to Anycast address is delivered to the closest interface (as defined by the routing protocols in use) identified as the Anycast address.
- Anycast address is syntactically indistinguishable from Unicast. Addresses because Anycast addresses are allocated from the Unicast address space
- Nodes to which the Anycast address is assigned must be explicitly configured to recognize that the address is an Anycast Address.
- Link to RFC:
→ RFC 2526

Anycast Address:

Syntax:

IPV4 Address: 2402::/64 Anycast

